

Study Report: Building LLM Powered Applications - Agentic AI Engineer

Written by Gagan Pasupuleti

Family: Agentic AI Engineer
Level: Intermediate
Chapters: 7
Source: Reference: Building LLM Powered Applications - Valentina Alto

Official sources and free libraries

- <https://www.oreilly.com/library/view/building-llm-powered/9781098152918/>

Summary

Study report for **Building LLM Powered Applications** by Valentina Alto (Intermediate level) mapped to the Agentic AI Engineer role. Production LLM apps: RAG, agents, guardrails, and scaling.

Chapters

Track: Book-Intro

Chapter 1: About Building LLM Powered Applications [Intermediate]

Chapter focus: About Building LLM Powered Applications** **(Level: Intermediate)**

This study report summarizes **Building LLM Powered Applications** by Valentina Alto for the Agentic AI Engineer role. The resource is rated Intermediate level. Production LLM apps: RAG, agents, guardrails, and scaling. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# Building LLM Powered Applications
# Author: Valentina Alto
# Role: Agentic AI Engineer
# Level: Intermediate
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is

A structured study report based on Building LLM Powered Applications.

When to use it

When learning Agentic AI Engineer skills at Intermediate level.

Where to use it

Agentic AI Engineer training paths and certification prep.

Real use example

A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Intermediate]

Chapter focus: Why This Book Matters for Your Role** *(Level: Intermediate)

Agentic AI Engineer professionals use ideas from Building LLM Powered Applications to solve real workplace problems. Production LLM apps: RAG, agents, guardrails, and scaling. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

Role: Agentic AI Engineer

Book focus: Production LLM apps: RAG, agents, guardrails, and scaling.

Recommended level: Intermediate

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is

Role-aligned learning connects theory to job tasks.

When to use it

During career planning and syllabus design.

Where to use it

Agentic AI Engineer bootcamps and CodeQuest teacher assignments.

Real use example

A teacher assigns this book report before a intermediate skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Intermediate]

Chapter focus: Key Topics Covered** *(Level: Intermediate)

The main topics in Building LLM Powered Applications include practical concepts described as: Production LLM apps: RAG, agents, guardrails, and scaling. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in Agentic AI Engineer jobs today.

Code Reference:

- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is

Topic maps turn a book into actionable learning objectives.

When to use it

Before reading and while building chapter summaries.

Where to use it

Self-study, flipped classroom, and revision.

Real use example

A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: LangGraph StateGraph Fundamentals [Intermediate]

Chapter focus: LangGraph StateGraph Fundamentals** *(Level: Intermediate)

LangGraph models agents as graphs: nodes mutate shared state; edges route conditionally. Supports cycles, persistence, and interrupts - unlike DAG-only chains.

Code Reference:

```
from langgraph.graph import StateGraph, END

graph = StateGraph(AgentState)
graph.add_node('plan', plan_node)
graph.add_node('act', act_node)
graph.add_edge('plan', 'act')
graph.add_conditional_edges('act', should_continue, {'continue': 'plan', 'end': END})
```

What it shows: conditional_edges implement loop-or-stop logic declaratively.

Topic guide

What it actually is

LangGraph is a framework for durable, stateful agent workflows.

When to use it

Production agents needing checkpoints and branching.

Where to use it

Support automation, coding agents, and ops runbooks.

Real use example

Graph resumes from checkpoint after server restart mid-workflow.

Chapter 5: Applied: MCP: Model Context Protocol [Intermediate]

Chapter focus: MCP: Model Context Protocol** *(Level: Intermediate)

MCP (Anthropic, 2024-2025) standardizes how hosts connect to tools/data via JSON-RPC. Servers expose resources, prompts, and tools - Cursor and Claude Desktop use MCP today.

Code Reference:

```
{
  "jsonrpc": "2.0",
  "method": "tools/call",
  "params": {"name": "query_db", "arguments": {"sql": "SELECT 1"}}
}
```

What it shows: tools/call invokes MCP server capability with typed args.

Topic guide

What it actually is

MCP is an open protocol for portable tool integrations across LLM hosts.

When to use it

Building reusable connectors instead of one-off agent hacks.

Where to use it

IDE assistants, enterprise copilots, and shared data access layers.

Real use example

Railway MCP server deploys services - agent uses same tool in Cursor and internal bot.

Chapter 6: Applied: Building an MCP Server [Intermediate]

Chapter focus: Building an MCP Server** *(Level: Intermediate)

Python MCP SDK: define @server.list_tools and @server.call_tool handlers. Run stdio for local

hosts or SSE for remote. Validate inputs; never expose raw shell.

Code Reference:

```
from mcp.server import Server
server = Server('docs-search')

@server.list_tools()
async def list_tools():
    return [Tool(name='search', description='Search internal wiki', inputSchema={...})]
```

What it shows: list_tools advertises capabilities; call_tool executes with validation.

Topic guide

What it actually is

Custom MCP servers package domain tools for any compatible host.

When to use it

Shared internal tools across Claude, Cursor, and custom agents.

Where to use it

Wiki search, Jira, and data catalog integrations.

Real use example

Docs MCP server powers both IDE and Slack agent with one codebase.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Intermediate]

Chapter focus: Study Plan and Practice** *(Level: Intermediate)

Finish Building LLM Powered Applications with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to Agentic AI Engineer. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes
Week 2: Exercises
Week 3: Mini project
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is

Structured study plans improve retention and portfolio outcomes.

When to use it

After finishing the key topics chapters.

Where to use it

CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a intermediate project screenshot.